BCS685 - Project Phase I

Review - 1

Multimodal Deepfake detection framework for real-time social media monitoring

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Introduction

Deepfake technology uses advanced **Artificial Intelligence (AI)** and **Deep Learning models** to generate highly realistic synthetic images and videos. While these technologies have useful applications in entertainment and media, they are increasingly misused to create **misleading or harmful content** on social media platforms.

The rapid spread of manipulated media has led to serious concerns such as:

- Online harassment and identity misuse
- Spread of misinformation and fake news
- Privacy violations and fraud
- Targeted attacks on vulnerable groups

Traditional deepfake detection methods often rely on **single-modality analysis** and struggle to detect increasingly sophisticated forgeries. Therefore, there is a need for **robust detection systems capable of identifying subtle artifacts across different signals.**

Problem Statement

- Deepfake technologies generate highly realistic synthetic media that can be misused for misinformation, identity fraud, and harassment.
- Existing detection systems mainly rely on **single-modality or spatial artifact analysis**, which fails to detect advanced deepfakes.
- Many current models suffer from **poor generalization across datasets and high computational cost**, limiting real-time deployment.
- There is a need for a **robust and scalable detection framework** capable of identifying subtle manipulation artifacts.
- The proposed system combines **frequency-domain artifact analysis and biological signal inconsistencies** to improve detection accuracy and reliability.

Objectives

- Develop a framework to **detect deepfake facial images** on social media platforms.
- Extract and analyze **frequency-domain artifacts** present in manipulated images.
- Identify **biological signal inconsistencies** in synthetic facial media.
- Improve **detection robustness and generalization** across different deepfake techniques.
- Detect deepfake facial media using deep learning techniques.
- Improve generalization across different deepfake generation methods.

Base Paper

Kou, Y., Li, P., Ma, H. et al. SFIAD: Deepfake detection through spatial-frequency feature integration and dynamic margin optimization. Artif Intell Rev 58, 217 (2025).

DOI: https://doi.org/10.1007/s10462-025-11225-7

Adaptive Texture and Spectrum Clue Mining for Generalizable Face Forgery Detection

Ref.	Author & Year	Title	Methodology	Key Findings	Limitations
[1]	**Authors:** Jiawei Liu, Jingyi Xie, Yang Wang, Zheng-Jun Zha **Year:** 2024	Adaptive Texture and Spectrum Clue Mining for Generalizable Face Forgery Detection	• The paper proposes an **Adaptive Texture and Spectrum Clue Mining framework** for detecting face forgery. • A deep CNN backbone extracts **high-level visual features** from facial images. • A **texture clue mining module** captures spatial artifacts such as abnormal skin textures and blending inconsistencies. • A **spectrum clue mining module** analyzes frequency-domain representations to detect hidden forgery artifacts. • The extracted spatial and spectral features are **adaptively fused for final real/fake classification**.	• Introduces a **dual-clue detection strategy** combining spatial texture and frequency spectrum information. • Enhances **generalization ability** to detect unseen deepfake generation methods. • Provides an adaptive mechanism to identify **subtle manipulation artifacts in facial images**. • Achieves improved detection performance across **multiple deepfake benchmark datasets**.	• Primarily designed for **image-based forgery detection**, without temporal analysis. • Frequency analysis may increase **computational complexity**. • Performance may degrade when **high-quality deepfakes contain minimal artifacts**.

A Unified Neural Framework for Real-Time Deepfake Detection Across Multimedia Modalities to Combat Misleading Content

Ref.	Author & Year	Title	Methodology	Key Findings	Limitations
[2]	**Authors:** Ayan Sar, Subhangi Sati, Tanupriya Choudhury, Purvika Joshi, Roohi Sille, K. Srihari **Year:** 2025	A Unified Neural Framework for Real-Time Deepfake Detection Across Multimedia Modalities to Combat Misleading Content	• The paper proposes a **unified deep learning framework** for detecting deepfakes across multimedia modalities. • The system processes **image, video, and audio inputs** to capture manipulation artifacts. • Spatial features are extracted from facial images and video frames using **deep neural networks**. • Audio signals are converted into **spectrogram representations** to detect speech inconsistencies. • Extracted multimodal features are **fused and passed to a classifier** for deepfake detection.	• Introduces a **multimodal deepfake detection framework** combining image, video, and audio data. • Enables **real-time deepfake detection** for multimedia platforms. • Improves robustness compared to **single-modality detection methods**.	• Requires availability of **multiple modalities** for optimal performance. • Multimodal processing leads to **higher computational cost**.

Advancing Generalization in Deepfake Detection: Supervised Contrastive Representation Learning With Dual-Stream Spatio-Temporal Features

Ref.	Author & Year	Title	Methodology	Key Findings	Limitations
[3]	**Authors:** Sangho Son, Wooju Kim **Year:** 2025	Advancing Generalization in Deepfake Detection: Supervised Contrastive Representation Learning With Dual-Stream Spatio-Temporal Features	• The paper proposes a **dual-stream spatio-temporal framework** for deepfake video detection. • A **spatial stream** extracts appearance-based facial features from individual frames using deep networks. • A **temporal stream** analyzes frame sequences to capture motion inconsistencies in manipulated videos. • Extracted spatial and temporal representations are **combined to form spatio-temporal feature embeddings**. • **Supervised contrastive learning** is used to enhance feature discrimination between real and fake samples.	• Introduces **supervised contrastive representation learning** to improve deepfake detection performance. • Proposes a **dual-stream architecture** capturing both spatial appearance and temporal motion patterns. • Enhances **cross-dataset generalization** for detecting unseen deepfake generation methods. • Demonstrates improved detection accuracy across **multiple deepfake video datasets**.	• Requires **video sequences**, making it unsuitable for single-image forgery detection. • Dual-stream architecture increases **training and computational complexity**. • Performance may depend on the **quality and diversity of training datasets**.

AV-Lip-Sync: Leveraging AV-HuBERT to Exploit Multimodal Inconsistency for Deepfake Detection of Frontal Face Videos

Ref.	Author & Year	Title	Methodology	Key Findings	Limitations
[4]	**Authors:** Sahibzada Adil Shahzad, Ammarah Hashmi, Yan-Tsung Peng, Yu Tsao, Hsin-Min Wang **Year:** 2025	AV-Lip-Sync: Leveraging AV-HuBERT to Exploit Multimodal Inconsistency for Deepfake Detection of Frontal Face Videos	• The paper proposes a **multimodal deepfake detection framework** based on audio–visual synchronization analysis. • Facial regions and **lip movement sequences** are extracted from video frames. • Audio signals are processed to obtain **speech feature representations**. • The **AV-HuBERT model** learns correlations between audio speech patterns and visual lip movements. • Lip-sync inconsistencies between audio and video are analyzed to classify **real and manipulated videos**.	• Introduces a **lip-sync based multimodal deepfake detection approach**. • Utilizes **AV-HuBERT architecture** to learn audio–visual speech relationships. • Detects deepfakes by identifying **audio–visual inconsistencies in lip synchronization**. • Demonstrates improved detection performance for **frontal face video deepfakes**.	• Requires **both audio and visual modalities** for accurate detection. • Primarily designed for **frontal face videos with visible lip movements**. • Performance may decrease when deepfakes achieve **highly accurate lip synchronization**.

Dual Frequency Branch Framework With Reconstructed Sliding Windows Attention for AI-Generated Image Detection

Ref.	Author & Year	Title	Methodology	Key Findings	Limitations
[5]	**Authors:** Jiazhen Yan , Ziqiang Li , Fan Wang, Ziwen He, Zhangjie Fu **Year:** 2025	Dual Frequency Branch Framework With Reconstructed Sliding Windows Attention for AI-Generated Image Detection	• The paper proposes a **dual-frequency branch framework** for detecting AI-generated images . • Input images are transformed into the **frequency domain** to reveal hidden spectral artifacts. • Two parallel branches extract **low-frequency structural features and high-frequency artifact patterns**. • A **Reconstructed Sliding Window Attention (RSWA) module** focuses on important local regions. • The extracted features are fused and passed to a classifier for **real vs AI-generated image detection**.	• Introduces a **dual-frequency analysis strategy** to detect subtle artifacts in generated images. • Utilizes **frequency-domain representations** to capture hidden manipulation patterns. • Proposes **RSWA attention mechanism** to enhance feature learning and artifact localization. • Demonstrates improved detection performance on **multiple AI-generated image datasets**.	• Primarily designed for **image-based forgery detection**, without temporal analysis. • Frequency transformation and dual branches increase **computational complexity**. • Performance may decrease for **high-quality generated images with minimal frequency artifacts**.

Dynamic Difference Learning With Spatio-Temporal Correlation for Deepfake Video Detection

Ref.	Author & Year	Title	Methodology	Key Findings	Limitations
[6]	**Authors:** Qilin Yin , Wei Lu , Member, Bin Li, Jiwu Huang **Year:** 2025	Dynamic Difference Learning With Spatio-Temporal Correlation for Deepfake Video Detection	• The paper proposes a **dynamic difference learning framework** for detecting deepfake videos. • Consecutive video frames are processed to compute **dynamic difference maps** highlighting motion changes. • A deep neural network learns **spatial features from facial regions** and temporal patterns from frame sequences. • Spatio-temporal correlation modeling captures **motion inconsistencies and abnormal facial movements**. • The learned features are aggregated and used to classify videos as **real or manipulated**.	• Introduces **dynamic difference learning** to capture subtle frame-to-frame inconsistencies. • Utilizes **spatio-temporal correlation modeling** for improved video-level deepfake detection. • Enhances detection accuracy by analyzing **temporal motion patterns in facial regions**. • Demonstrates improved performance on **multiple deepfake video benchmark datasets**.	• Requires **video sequences**, making it unsuitable for single image detection. • Temporal modeling increases **computational and memory requirements**. • Performance may reduce for **high-quality deepfake videos with realistic motion patterns**.

ISTVT: Interpretable Spatial-Temporal Video Transformer for Deepfake Detection

Ref.	Author & Year	Title	Methodology	Key Findings	Limitations
[7]	**Authors:** Cairong Zhao, Chutian Wang, Guosheng Hu, Haonan Chen, Chun Liu, Jinhui Tang **Year:** 2023	ISTVT: Interpretable Spatial-Temporal Video Transformer for Deepfake Detection	• The paper proposes an **Interpretable Spatial-Temporal Video Transformer (ISTVT)** for deepfake detection. • Video frames are extracted and facial regions are processed to obtain **spatial feature representations**. • A **video transformer architecture** models long-range relationships between frames. • Self-attention mechanisms capture **spatial textures and temporal motion patterns**. • The learned representations are used to classify videos as **real or deepfake**.	• Introduces a **spatial-temporal transformer architecture** for deepfake video detection. • Utilizes **self-attention mechanisms** to capture long-range dependencies across video frames. • Provides **interpretable attention maps** highlighting manipulated facial regions. • Achieves improved detection performance on **benchmark deepfake video datasets**.	• Transformer-based models require **large training datasets and high computational resources**. • Processing long video sequences increases **memory and processing overhead**. • Detection performance may depend on **video quality and dataset diversity**.

Ref.	Author & Year	Title	Methodology	Key Findings	Limitations
[8]	**Authors:** B. G. Deepa, C. K. Lokesh, D. Umamaheswari, B. Ayshwarya, P. V. Yethish, K. P. Suhaas Yi Kou, Peng Li, Hongjiang Ma, Jiliu Zhou, Zhan Ao Huang, Xiaojie Li **Year:** 2026	An Enhanced Deep Learning Framework for DeepFake Detection using EfficientNet-B3: Comparative Evaluation of Deep and Machine Learning Techniques	• Preprocess images through resizing, normalization, and data augmentation. • Extract deep features using a **pretrained EfficientNet-B3 model**. • Apply transfer learning to adapt the model for deepfake detection. • Use a custom classification head with fully connected layers and dropout. • Train the model using **Binary Cross-Entropy loss and Adam optimizer**.	• EfficientNet-B3 achieved the highest performance with approximately **97.95% detection accuracy**. • Deep learning approaches outperform traditional machine learning methods for deepfake detection tasks. • Transfer learning improves model performance while reducing training time and required data. • The model effectively captures complex manipulation artifacts present in deepfake images.	• The model mainly focuses on **image-based deepfake detection rather than video analysis**. • Limited dataset diversity may affect the model's generalization ability. • The framework was not evaluated against adversarial attacks or heavily compressed media.

Deepfake Detection using Convolutional Vision Transformers and Convolutional Neural Network

Ref.	Author & Year	Title	Methodology	Key Findings	Limitations
[9]	**Authors:** Ahmed Hatem Soudy, Omnia Sayed, Hala Tag-Elser, Rewaa Ragab, Sohaila Mohsen, Tarek Mostafa, Amr A. Abohany, Salwa O. Slim **Year:** 2024	Deepfake Detection using Convolutional Vision Transformers and Convolutional Neural Networks	• The system extracts frames from input videos and performs face detection using **MTCNN**. • Facial regions including the **entire face, eyes, and nose** are segmented for detailed analysis. • Separate **CNN models** are used to analyze eye and nose regions. • A hybrid **CNN + Vision Transformer architecture** is used to analyze the full face region and capture global contextual relationships. • Predictions from the three models are combined using a **majority voting ensemble mechanism** to determine the final classification.	• Analyzing multiple facial regions improves deepfake detection accuracy. • CNN models effectively capture local spatial artifacts while transformers capture global image relationships. • The ensemble majority voting approach improves reliability and robustness of predictions. • The combined system achieved approximately **97% overall deepfake detection accuracy** on benchmark datasets.	• The framework requires significant computational resources due to multiple models. • The system mainly focuses on facial regions and may miss manipulations occurring outside the face. • Performance may decrease when encountering unseen deepfake generation techniques.

Frequency Domain Filtered Residual Network for Deepfake Detection

Ref.	Author & Year	Title	Methodology	Key Findings	Limitations
[10]	**Authors:** Bo Wang, Xiaohan Wu, Yeling Tang, Yanyan Ma, Zihao Shan, Fei Wei **Year:** 2023	Frequency Domain Filtered Residual Network for Deepfake Detection	• The input RGB images are first converted into grayscale for frequency analysis. • A **Haar Wavelet Transform** is applied to extract frequency components from the image. • A **residual map** representing mid-high frequency information is generated by subtracting low-frequency components. • The residual frequency map is concatenated with the original RGB image channels to form a fused feature representation. • The combined features are processed using an **Xception CNN network** for deepfake classification.	• Frequency-domain features reveal hidden artifacts produced during deepfake generation. • Combining RGB features with frequency residual maps improves detection robustness. • The proposed method achieved **over 98% accuracy for uncompressed deepfake images**. • The model performs well across datasets such as **FaceForensics++, Celeb-DF, and UADFV**.	• Detection accuracy decreases for certain manipulation techniques such as Face2Face. • Performance drops when images are extremely compressed. • The method focuses mainly on image-level detection and does not analyze temporal relationships in videos.

Audio–Visual Synchronization and Lip Movement Analysis for Real-Time Deepfake Detection

Ref.	Author & Year	Title	Methodology	Key Findings	Limitations
[11]	**Authors:** Muhammad Javed, Zhaohui Zhang, Fida Hussain Dahri, Asif Ali Laghari, Martin Krajčík, Ahmad Almadhor **Year:** 2025	Audio–Visual Synchronization and Lip Movement Analysis for Real-Time Deepfake Detection	• Video frames are extracted and lip regions are detected using face detection techniques. • Spatial lip features are extracted using **CNN and Temporal Convolutional Networks (TCN)**. • Audio signals are processed using **Short-Time Fourier Transform and MFCC feature extraction**. • Temporal dependencies between audio and visual features are modeled using **BiLSTM networks**. • The system aligns lip movements and speech signals using **Dynamic Time Warping (DTW)** and performs multimodal classification.	• Combining audio and visual information improves deepfake detection performance. • Lip-speech synchronization inconsistencies are strong indicators of manipulated videos. • The framework achieved high accuracy across datasets such as **FakeAVCeleb, AV-Deepfake1M, TVIL, and LAV-DF**. • The system supports near **real-time deepfake detection**.	• Detection accuracy decreases for videos with very low resolution or heavy noise. • Audio quality degradation can affect synchronization analysis . • Multimodal processing increases computational requirements.

Enhancing Multimodal Deepfake Detection with Local–Global Feature Integration and Diffusion Models

Ref.	Author & Year	Title	Methodology	Key Findings	Limitations
[12]	**Authors:** Muhammad Javed, Zhaohui Zhang, Fida Hussain Dahri, Asif Ali Laghari, Martin Krajčík, Ahmad Almadhor **Year:** 2025	Enhancing Multimodal Deepfake Detection with Local–Global Feature Integration and Diffusion Models	• Input video frames are first processed using **diffusion models to enhance data quality and remove noise**. • Local facial features such as lips and eyes are extracted using **CNN networks**. • Global contextual relationships across frames are captured using **Vision Transformers (ViT)**. • Audio features are extracted using **MFCC feature extraction and BiLSTM networks**. • The audio and visual features are aligned using **Dynamic Time Warping** and fused for classification..	• Combining CNN and Vision Transformers improves both local and global feature representation. • Diffusion preprocessing enhances feature quality and improves detection robustness. • Multimodal analysis improves detection accuracy compared to unimodal approaches. • The framework achieves **state-of-the-art performance across multiple deepfake datasets**.	• The integration of transformers and diffusion models increases computational complexity. • Real-time implementation requires powerful hardware resources. • Performance may decrease for extremely low-quality or highly compressed videos.

DFFMD: A Deepfake Face Mask Dataset for Infectious Disease Era With Deepfake Detection Algorithms

Ref.	Author & Year	Title	Methodology	Key Findings	Limitations
[13]	**Authors:** Muhammad Javed, Zhaohui Zhang, Fida Hussain Dahri, Asif Ali Laghari, Martin Krajčík, Ahmad Almadhor **Year:** 2023	DFFMD: A Deepfake Face Mask Dataset for Infectious Disease Era With Deepfake Detection Algorithms	• A new dataset called **Deepfake Face Mask Dataset (DFFMD)** is created containing real and fake masked videos. • Fake videos are generated using the **First Order Motion Model (FOMM)**. • Video frames are extracted and facial regions are detected and cropped for preprocessing. • Deep learning models including **CNN, VGG19, and Inception-ResNet-v2** are trained for detection. • Model performance is evaluated using metrics such as **accuracy, precision, recall, F1 score, and AUC**.	• Transfer learning models outperform basic CNN architectures for masked-face deepfake detection. • **Inception-ResNet-v2 achieved the highest accuracy of approximately 99.81%**. • The proposed dataset enables research on deepfake detection in scenarios where faces are partially covered. • Deep learning models remain effective even when facial features are partially hidden by masks.	• The dataset size is relatively small compared to large deepfake datasets. • Fake videos generated using FOMM may have medium visual quality. • The generated deepfake videos lack original audio, limiting multimodal analysis.

CNN-Based Metrics for Performance Evaluation of Generative Adversarial Networks

Ref.	Author & Year	Title	Methodology	Key Findings	Limitations
[14]	**Authors:** Adarsh Prasad Behera, Satya Prakash, Siddhant Khanna, Shivangi Nigam, Shekhar Verma **Year:** 2024	CNN-Based Metrics for Performance Evaluation of Generative Adversarial Networks	• Proposed two CNN-based evaluation metrics: **Classification Score (CS)** and **Distribution Score (DS)**. • A CNN classifier is trained on real images and used to evaluate GAN-generated images. • **Direct Classification, Indirect Classification, and Blind Classification** strategies are used. • Performance of different GAN models is evaluated using these metrics.	• Proposed metrics effectively detect **mode collapse and overfitting** in GANs. • Provides more reliable GAN evaluation than traditional metrics. • Helps compare performance of different generative models.	• Performance depends on the **quality of the CNN classifier**. • Metrics may not fully capture **visual realism perceived by humans**.

Div-Df: A Diverse Manipulation Deepfake Video Dataset

Ref.	Author & Year	Title	Methodology	Key Findings	Limitations
[15]	**Authors:** Deepak Dagar, Dinesh Kumar Vishwakarma **Year:** 2023	Div-DF: A Diverse Manipulation Deepfake Video Dataset	• Introduced a new dataset called **Div-DF** containing real and manipulated videos. • Includes multiple deepfake manipulation techniques: **face swap, facial reenactment, and lip sync**. • Fake videos are generated using **FSGAN and Wav2Lip models**. • Benchmark experiments conducted using several deepfake detection models.	• Dataset increases **manipulation diversity** compared to existing datasets. • Improves evaluation of deepfake detection algorithms. • Shows that many existing detectors struggle with **diverse manipulations**.	• Dataset size is **smaller than large datasets like DFDC**. • Limited number of identities and manipulation techniques.

Exploiting Facial Relationships and Feature Aggregation for Multi-Face Forgery Detection

Ref.	Author & Year	Title	Methodology	Key Findings	Limitations
[16]	**Authors:** Yinan Zhao, Xiaokang Yang, Yong Xu, Jingdong Wang **Year:** 2024	Exploiting Facial Relationships and Feature Aggregation for Multi-Face Forgery Detection	• Proposed the **FILTER framework** for detecting deepfakes in multi-face images. • Introduced **Multi-Face Relationship Learning (MRL)** to capture relationships between faces. • Used **Global Feature Aggregation (GFA)** to combine facial and scene features. • Applied **contrastive learning** to improve feature separation.	• Learning relationships between faces improves detection accuracy. • Combining **local facial features and global image context** enhances performance. • Achieves strong performance on multi-face deepfake datasets.	• Performance decreases when **many faces appear in one image**. • Requires accurate **face detection preprocessing**.

An Attention Fusion-Based Multimodal Autoencoder Deepfake Detection Framework

Ref.	Author & Year	Title	Methodology	Key Findings	Limitations
[17]	**Authors:** Tong Hu, Wei Liu, Jie Yang, Yaqi Fan, Yuanyuan Qiao **Year:** 2025	An Attention Fusion-Based Multimodal Autoencoder Deepfake Detection Framework	• Proposed a **multimodal deepfake detection framework** using audio and visual features. • Visual features extracted from video frames using CNN models. • Audio features extracted using **spectrogram and MFCC representations**. • **Attention-based fusion module** combines audio and visual information. • Multimodal **autoencoder reconstructs features to detect anomalies**.	• Multimodal detection improves accuracy compared to visual-only methods. • Attention fusion effectively balances audio and visual features. • Detects **audio-visual inconsistencies in deepfake videos**.	• Requires videos with **audio signals**. • Higher computational complexity due to multimodal processing.

AVT2-DWF: Improving Deepfake Detection With Audio-Visual Fusion and Dynamic Weighting Strategies

Ref.	Author & Year	Title	Methodology	Key Findings	Limitations
[18]	**Authors:** Rui Wang, Dengpan Ye, Long Tang, Yunming Zhang, Jiacheng Deng **Year:** 2024	AVT2-DWF: Improving Deepfake Detection With Audio-Visual Fusion and Dynamic Weighting Strategies	• Proposed **Audio-Visual Transformer (AVT)** for deepfake detection. • Extracted visual features from video frames using CNN models. • Extracted audio features using **Mel spectrogram representations**. • Transformer models **temporal relationships between audio and visual signals**. • Introduced **Dynamic Weight Fusion (DWF)** to adaptively weight modalities.	• Transformer improves modeling of **temporal audio-visual relationships**. • Dynamic weighting improves robustness when one modality is noisy. • Achieves higher detection accuracy compared to traditional multimodal methods.	• Requires both **audio and visual data**. • Transformer models require **high computational resources**.

Face Forgery Detection With CLIPEnhanced Multi-Encoder Distillation

Ref.	Author & Year	Title	Methodology	Key Findings	Limitations
[19]	**Authors:** Chunlei Peng, Tianzhe Yan, Decheng Liu, Nannan Wang, Ruimin Hu, and Xinbo Gao **Year:** 2023	Face Forgery Detection With CLIP-Enhanced Multi-Encoder Distillation	• Proposed a **CLIP-enhanced deepfake detection framework**. • Multiple encoders extract different types of visual features. • **CLIP visual encoder provides semantic representations**. • Teacher–student **knowledge distillation framework** transfers knowledge to a lightweight model. • Feature fusion combines outputs from multiple encoders.	• CLIP improves **cross-dataset generalization**. • Multi-encoder architecture captures **diverse forgery patterns**. • Knowledge distillation reduces model complexity while maintaining accuracy.	• Training teacher model requires **high computational resources**. • Performance depends on **pretrained CLIP model quality**.

Gaps Identification

• Many existing deepfake detection methods rely mainly on **spatial visual artifacts**, which are less effective for detecting advanced deepfakes.

• Several approaches focus on **single-modality detection** (image, video, or audio) instead of integrating multiple signals.

• Current models often show **poor generalization across unseen datasets and new deepfake generation techniques**.

• Some deepfake detectors require **high computational resources**, limiting real-time deployment on social media platforms.

• Limited research combines **frequency-domain artifacts with biological signal inconsistencies** for more reliable detection.

Timeline

- **Phase 1 – Problem Analysis & Literature Survey**
  Study deepfake detection techniques and analyze existing research papers.
- **Phase 2 – Dataset Collection & Preprocessing**
  Collect deepfake datasets and perform image/video preprocessing.
- **Phase 3 – Model Design & Feature Extraction**
  Design the detection framework and extract frequency-domain and biological signal features.
- **Phase 4 – Model Implementation & Training**
  Implement the deep learning model and train it using the prepared dataset.
- **Phase 5 – Testing & Performance Evaluation**
  Evaluate model performance using accuracy, precision, recall, and other metrics.
- **Phase 6 – Result Analysis & Project Documentation**
  Analyze results, optimize the model, and prepare the final project report and presentation.

Software / Hardware Requirements

Software Requirements

- Programming Language: **Python**
- Development Environment: **Jupyter Notebook / Google Colab / VS Code**
- Libraries & Frameworks: **TensorFlow / PyTorch, OpenCV, NumPy, Pandas, Matplotlib**
- Deep Learning Tools: **Scikit-learn, Torchvision / Keras**
- Dataset Handling: **FaceForensics++, Celeb-DF, DFDC datasets**

Hardware Requirements

- Processor: **Intel i5 / i7 or equivalent**
- RAM: **16 GB or higher**
- Storage: **Minimum 50 GB free space**
- GPU (optional but recommended): **NVIDIA GPU with CUDA support**
- System: **Laptop / Desktop with internet connectivity**

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